SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I V THANISHA | 192526060 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thf

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	6
Experimental Design	30	5
Use of Tools	20	5
Analysis of Results	15	5
Documentation	10	5

Total Marks 26 /30

Signature of the Faculty

TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretatio	Basic analysis with limited insights	Poor or incorrect analysis of results

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Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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CO1 – AT3 TECHNICAL PRESENATION
EDGE COMPUTING AND RELIABLE DATA TRANSMISSION

Name: V Thanisha

Registration Number: 1965260160

Course: Computer Networks

Course Code: CSA0709

Faculty Name: Dr.R Shankar

Introduction

- Edge computing is a decentralized computing framework that brings data processing, analysis, and storage physically closer to the data source (such as IoT devices, local gateways, or on-premises servers) rather than sending all raw data to a distant, centralized cloud or data center.

- Reliable data transmission ensures that information collected and processed at the edge reaches its destination (cloud, control center, or peer device) without loss, corruption, or excessive delay.

How Edge Computing Processes Data Near Its Source

- **Local Data Capture & Collection:** Smart devices, cameras, or machinery sensors collect continuous streams of raw data at the point of origin.

- **Filtering and Pre-Processing:** Instead of sending all raw data over the network, local edge hardware (like gateways or micro-servers) cleans, compresses, and filters the information.

- **Immediate Local Decision-Making:** Edge nodes run local algorithms, rules, or machine learning models to analyze time-sensitive data and trigger instant actions in milliseconds (such as braking in an autonomous vehicle or shutting down a factory valve).

- **Selective Cloud Transmission:** Only summarized, critical, or long-term storage data is sent upward to centralized cloud or enterprise data centers.

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How Edge Computing Enables Reliable Data Transmission

- **Reduced Network Dependence:** Because heavy computation happens locally, systems do not rely on a continuous, high-speed connection to a central server.

- **Resilience During Outages:** Edge devices can operate offline or during intermittent internet drops, storing data locally until connectivity is restored.

- **Bandwidth Optimization:** Filtering out redundant or low-priority data locally prevents network congestion and lowers transit costs over wide-area networks

- **Enhanced Data Security & Privacy:** Processing sensitive information on-site minimizes exposure across public or long-distance transmission lines and helps comply with regional data laws.

Challenges in Edge Computing and Reliable Data Transmission

1. Network Failures

Network connections may be interrupted due to hardware faults or weak connectivity. Edge devices may temporarily lose communication with servers.

Solution: Use backup connections, redundant paths, and fault-tolerant systems.

2. Latency

Latency is the time taken for data to travel from the source to the destination. Network congestion and slow processing can delay data delivery.

Solution: Process data near the source using edge computing and optimize network traffic.

3. Packet Loss

Packet loss occurs when some data packets fail to reach their destination. It can happen because of network congestion, interference, or unstable connections.

Solution: Use error detection, acknowledgments, retransmission, and suitable error-correction techniques

Reliable Data Transmission

Reliable data transmission is the process of sending data from a source to a destination accurately, completely, and with minimal errors or delays.

Key techniques for reliable transmission

1. Error Detection
Detects corrupted data using methods such as checksums and cyclic redundancy checks (CRC).

2. Retransmission
Lost or corrupted data packets are sent again to ensure successful delivery.

3. Acknowledgment
The receiver confirms that data has been received successfully.

4. Redundancy and Fault Tolerance
Alternative communication paths or backup components help maintain data delivery when a link or device fails

Applications of Edge Computing and Reliable Data Transmission

1. Internet of Things (IoT)

Connects smart devices and sensors to edge servers. Enables real-time data processing and monitoring. Reliable transmission helps reduce data loss between devices.

Example: Smart home devices monitor temperature, lighting, and energy usage.

2. Smart Cities

Supports intelligent traffic management and smart street lighting. Processes data from cameras and sensors near the source. Reliable communication helps coordinate city services efficiently.

Example: Traffic sensors send real-time vehicle data to edge servers to adjust traffic signals.

3. Healthcare

Processes patient monitoring data close to medical devices. Reduces delays in delivering important health information. Reliable transmission helps maintain accurate patient records and timely alerts.

Example: Wearable sensors transmit heart rate and other vital signs to a hospital monitoring system.

Future Scope of Edge Computing and Reliable Data Transmission

1. 5G and 6G Networks

5G enables high-speed communication with low latency and supports large numbers of connected devices. 6G is expected to provide even higher performance, intelligent connectivity, and advanced communication capabilities.These networks can support real-time applications such as autonomous vehicles, smart cities, and remote monitoring.

Example: Connected vehicles exchange information quickly to improve road safety.

2. AI-Based Networking

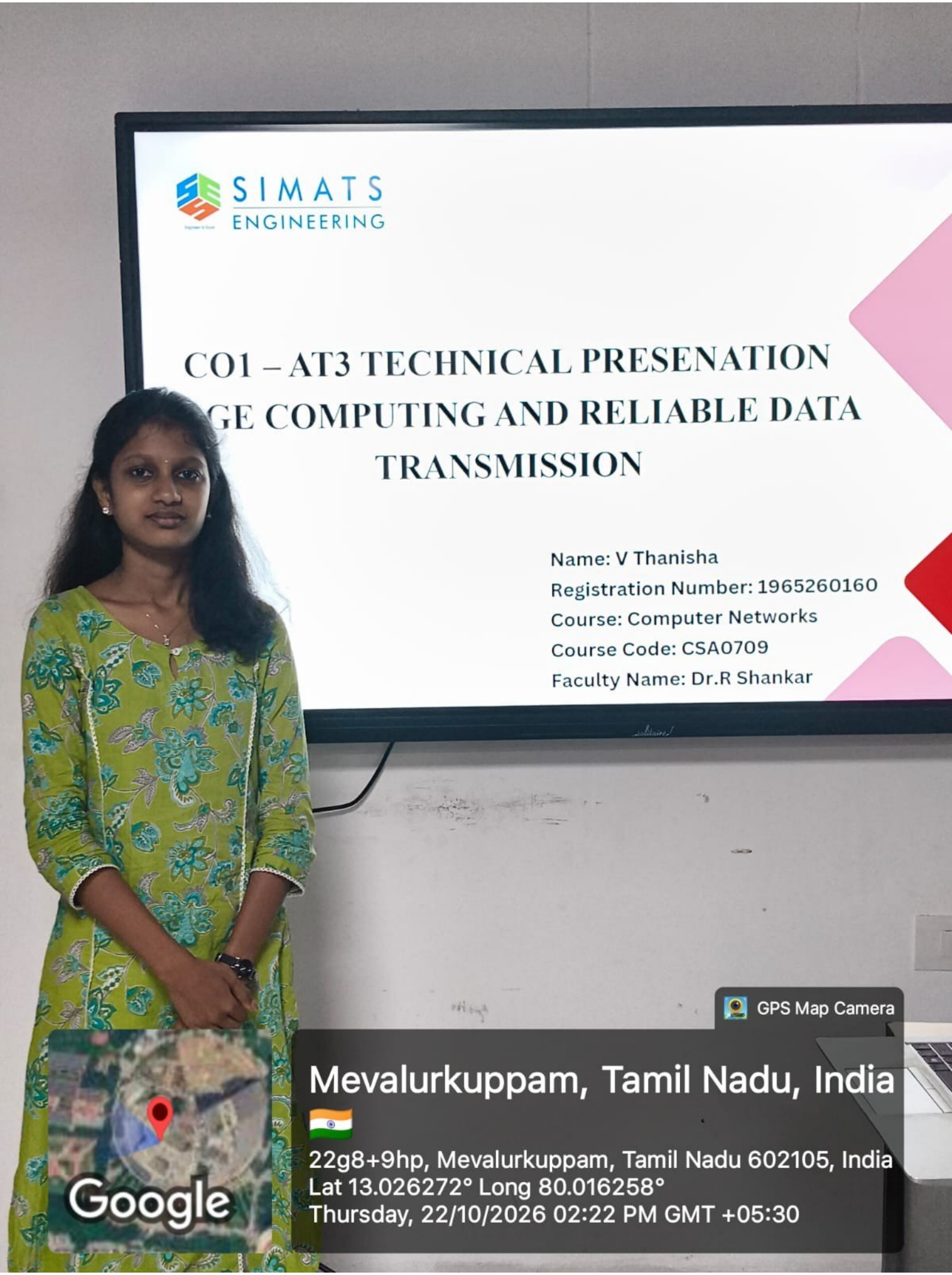
Artificial Intelligence (AI) can monitor network performance and detect unusual behavior.

AI can predict network congestion and optimize data routing. It can help detect failures and improve resource allocation.

Example: An AI-based network selects a less congested route to transmit data efficiently.

Conclusion

Edge computing and reliable data transmission play an important role in modern communication systems. Edge computing processes data closer to its source, reducing latency and network traffic. Reliable data transmission ensures that information reaches its destination accurately and efficiently. Techniques such as error detection, retransmission, redundancy, and fault tolerance help minimize data loss and improve system reliability. These technologies are widely used in IoT, smart cities, healthcare, and autonomous systems. In the future, 5G/6G networks, AI-based networking, and improved edge infrastructure will further enhance communication speed, security, and reliability. Overall, edge computing combined with reliable data transmission provides a faster, smarter, and more dependable foundation for future digital technologies.


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GE COMPUTING AND RELIABLE DATA
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